Trader Behavior vs Market Sentiment Analysis

Junior Data Scientist – Trader Behavior Insights

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1. Introduction

Cryptocurrency markets are highly volatile and often influenced by collective market sentiment, commonly described using Fear and Greed indicators. While sentiment indices are frequently referenced by traders, there is limited empirical analysis connecting actual trader execution behavior with prevailing market sentiment.

This project analyzes the relationship between trader performance, risk behavior, and market sentiment by combining historical trader-level execution data with the Bitcoin Fear–Greed sentiment index. The objective is to uncover meaningful patterns that can support smarter, sentiment-aware trading strategies.

2. Problem Statement

Trader decision-making in crypto markets is often driven by psychological factors such as fear and greed. However, it remains unclear how these sentiments translate into real trading outcomes, such as profitability, risk exposure, and consistency across different market conditions.

The problem addressed in this project is to determine:

- Whether trader performance differs during Fear vs Greed market regimes
- How trader risk behavior changes with sentiment
- Whether certain traders consistently perform well regardless of sentiment

3. Objective

The primary objective of this analysis is to:

Understand how trader behavior and performance align or diverge from market sentiment (Fear vs Greed) and identify actionable patterns that can inform smarter trading strategies.

Specific goals include:

- Aligning trader execution data with daily market sentiment
- Evaluating profitability and win rates across sentiment regimes
- Analyzing risk exposure using valid proxies
- Identifying sentiment-resilient traders

4. Datasets Used

4.1 Bitcoin Market Sentiment Dataset

- **Columns:** Date, Classification (Fear / Greed)
- Represents daily market sentiment for Bitcoin

4.2 Historical Trader Data (Hyperliquid)

- **Key Columns:**
    - account
    - symbol
    - execution price
    - size
    - side
    - time
    - closedPnL

Note: The dataset does not contain explicit leverage information.

5. Data Preprocessing

- Converted trade timestamps to datetime format
- Extracted trade dates to align with daily sentiment
- Removed records with missing critical values
- Standardized sentiment labels for consistency
- Merged trader data with sentiment data using trade date

This ensured that each trade was analyzed under the correct sentiment condition.

6. Feature Engineering

Since leverage data was unavailable, risk was approximated using position value, defined as:

Position Value = |Trade Size| × Execution Price

Additional features created:

- Profitability Flag: Whether a trade resulted in positive PnL
- PnL Ratio: Closed PnL normalized by position value

These features allowed meaningful analysis of risk and performance without introducing assumptions.

7. Exploratory Data Analysis (EDA)

The analysis explored:

- Trade frequency under Fear vs Greed
- Distribution of closed PnL across sentiments
- Risk exposure differences using position value
- Normalized profitability trends

Visualizations revealed noticeable behavioral differences between sentiment regimes, particularly in trade aggressiveness and risk exposure.

8. Aggregated Sentiment-Level Analysis

Key metrics were computed for each sentiment category:

- Average and median PnL
- Win rate
- Average position value (risk proxy)
- Trade count

This enabled direct comparison of trader behavior and outcomes under Fear and Greed conditions.

9. Statistical Testing

A Welch's t-test was conducted to test whether the difference in PnL between Fear and Greed periods was statistically significant.

This step ensured that observed differences were data-driven and not based on visual intuition alone.

10. Trader-Level Behavioral Analysis

Trades were aggregated at the trader + sentiment level to analyze:

- Total profitability per sentiment
- Average risk exposure
- Trading frequency

This helped identify:

- Traders who perform well only under specific sentiments
- Traders who remain profitable across both Fear and Greed regimes

11. Key Insights

- Trader performance varies meaningfully across Fear and Greed periods
- Risk exposure (position value) is generally higher during Greed phases
- Greedy markets show more aggressive trading behavior
- Some traders demonstrate consistent profitability regardless of sentiment

12. Limitations

- Leverage data was unavailable, requiring the use of position value as a proxy
- Analysis is limited to historical data and does not imply causality
- Sentiment is measured at a daily level, which may not capture intraday shifts

13. Conclusion

This project demonstrates that market sentiment has a measurable relationship with trader behavior and performance. By aligning real execution data with sentiment indicators, the analysis provides actionable insights that can support sentiment-aware risk management and trading strategies.

The approach is transparent, statistically validated, and grounded in real-world data, making it suitable for further extension in live trading intelligence systems.

14. Future Work

- Incorporate leverage or margin data if available
- Extend analysis to intraday sentiment signals
- Cluster traders based on behavior profiles
- Build predictive models for sentiment-driven risk alerts

Declaration

All analyses were performed using the provided datasets without introducing fabricated variables. Assumptions and proxies used are clearly documented.